Category Theory for Mathematical Physics — Exercise Sheet 7

Braiding: hexagons, Yang–Baxter and topological spin

Variant: Hints and answers

Conventions as in Lecture 7: the crossing $c_{X,Y}$ takes the strand from bottom left to top right over the other; $c_{a,b}\psi_c^{ab} = R_c^{ab}\,\psi_c^{ba}$ (Equation (7.5)); the hexagon equations are the two families of Theorem 7.5, with labels (a, b, c, d, e, g). The F-symbols are those of Lecture 6 (Fibonacci $1, \tau$; Ising $1, \sigma, \psi$, with the bare ψ Ising's fermion). P is the flip of two tensor factors. In Exercise 4, $\mathbb{Z}_2\times\mathbb{Z}_2$ is written additively, with the house names $1 = (0,0)$, $e = (1,0)$, $m = (0,1)$, $\varepsilon = (1,1)$ (here e is an anyon and ε the toric-code fermion); elsewhere $\mathbb{Z}_2 = \{1, s\}$ is multiplicative. $\varphi = (1+\sqrt{5})/2$. The braid generators σ_i never meet the Ising anyon σ; in Exercise 5 the unknown braiding is $c_{s,s} = z$, to keep c for a hexagon label; in Exercise 6 the unknown $R_\sigma^{\sigma\psi}$ is named y (A is reserved for $q = A^2$).

Exercise 1 (Braids and permutations). (i) Show that $\sigma_i \mapsto (i\ i{+}1)$ defines a surjective homomorphism $B_n \to S_n$, and $\sigma_i \mapsto 1$ a homomorphism $B_n \to \mathbb{Z}$ (with $\mathbb{Z}$ additive). Conclude that for $n \geq 2$, σ_1 has infinite order in B_n, while its image in S_n has order 2.

(ii) For $n = 3$ draw $\sigma_1\sigma_2\sigma_1$, $\sigma_2\sigma_1\sigma_2$, and σ_1^2. Which drawing shows why $\sigma_1^2 \neq 1$?

Hint. (i) A homomorphism out of B_n is fixed by the images of the generators, once they satisfy the relations of § 7.1.

Answer. (i) The transpositions satisfy both relations and generate S_n; both sides of each relation have the same number of letters; $\sigma_1^k \mapsto k \neq 0$ for $k \neq 0$. (ii) σ_1^2: one strand winds once around the other, which no planar move undoes.

Exercise 2 (Two forms of Yang–Baxter). (i) Draw both sides of the braid relation of Theorem 7.4 for strands X, Y, Z, and say which steps of its proof use the first hexagon and which use naturality.

(ii) Let R act on $V \otimes V$, let $c = P \circ R$, and write R_{12}, R_{13}, R_{23} and c_{12}, c_{23} for the operators on $V^{\otimes 3}$. Show that

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23} \iff R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

(iii) The braiding of sVec on $V = \mathbb{C}^{1|1}$ is $c(v \otimes w) = (-1)^{|v||w|} w \otimes v$. Find its R, check both forms, and check that $c^2 = 1$.

Hint. (ii) Push every P to the left with $P_{12}X_{23}P_{12} = X_{13}$ and its relatives, and compare the permutations that come out.

Answer. (ii) $c_{12}c_{23}c_{12} = P_{12}P_{23}P_{12}\,R_{23}R_{13}R_{12}$ and $c_{23}c_{12}c_{23} = P_{23}P_{12}P_{23}\,R_{12}R_{13}R_{23}$, and $P_{12}P_{23}P_{12} = P_{23}P_{12}P_{23}$. (iii) $R = \mathrm{diag}(1, 1, 1, -1)$ in the basis $00, 01, 10, 11$; diagonal matrices commute, and $c^2(v \otimes w) = (-1)^{2|v||w|} v \otimes w$.

Exercise 3 (B_3 on three τ's). Take Fibonacci with $R_1^{\tau\tau} = e^{-4\pi i/5}$ and $R_\tau^{\tau\tau} = e^{3\pi i/5}$ (Theorem 7.6), and $F = F_\tau^{\tau\tau\tau}$ of Theorem 6.6.

(i) On $V_\tau^{\tau\tau\tau}$, with the basis of left trees L_1, L_τ, show that σ_1 acts by $M_1 = \mathrm{diag}(R_1^{\tau\tau}, R_\tau^{\tau\tau})$ and σ_2 by $M_2 = FM_1F$.

(ii) Show that the braid relation $M_1M_2M_1 = M_2M_1M_2$ is equivalent to $(FM_1)^3$ commuting with F, and check that $(FM_1)^3$ is a multiple of the identity.

(iii) How do σ_1 and σ_2 act on $V_1^{\tau\tau\tau}$?

Hint. (i) σ_2 acts diagonally on the right trees; use the F-move and its inverse (Equation (6.1), Equation (6.3)). (ii) $F^2 = 1$.

Answer. (i) σ_1 crosses the first two strands of L_e, giving $R_e^{\tau\tau}$; for σ_2, $L \to R$ by F, then $R_f^{\tau\tau}$, then back by $F^{-1} = F$. (ii) $(FM_1)^3 = e^{-4\pi i/5}\,1$. (iii) Both by $R_\tau^{\tau\tau}$: the only tree has inner label τ.

Exercise 4 (The toric code as a braided category). Take $\mathrm{Vec}_{\mathbb{Z}_2\times\mathbb{Z}_2}$ with trivial associator, and a braiding acting on simple objects by numbers, $c_{x,y} = \beta(x,y)\,\mathrm{id}$, with $\beta(x,0) = \beta(0,y) = 1$.

(i) Show that both families of hexagon equations hold exactly when β is a *bicharacter*: $\beta(x+y,z) = \beta(x,z)\beta(y,z)$ and $\beta(x,y+z) = \beta(x,y)\beta(x,z)$.

(ii) Let $\beta(x,y) = (-1)^{x_1y_2}$. Compute the spins θ_x of $1, e, m, \varepsilon$ and the table of double braidings $\beta(x,y)\beta(y,x)$. Is $\beta(x,y) = \beta(y,x)$? Is the braiding symmetric in the sense of Definition 7.2?

(iii) Show that $\beta'(x,y) = (-1)^{x_2y_1}$ is the gauge transform of β by $u(x,y) = \beta(x,y)$, which keeps the trivial associator.

Hint. (i) Every F-symbol is 1, and each sum has one term. (iii) A gauge multiplies R^{xy} by $u(y,x)/u(x,y)$, and the F-symbol ω by δu (Theorem 6.4).

Answer. (i) The first family reads $\beta(z,x)\beta(z,y) = \beta(z,x+y)$, the second $\beta(x,z)\beta(y,z) = \beta(x+y,z)$. (ii) $\theta = 1, 1, 1, -1$; the double braiding is -1 for the pairs $\{e,m\}$, $\{e,\varepsilon\}$, $\{m,\varepsilon\}$ and 1 otherwise; $\beta(e,m) = -1 \neq \beta(m,e) = 1$, and the braiding is not symmetric: the double braiding of e and m is -1. (iii) $\beta(x,y)\,\beta(y,x)/\beta(x,y) = \beta(y,x) = \beta'(x,y)$, and $\delta\beta = 1$ for a bicharacter.

Exercise 5 (Spins: the semion and Fibonacci). (i) For Vec_G^ω on $\mathbb{Z}_2$ with $\omega(s,s,s) = -1$ and $c_{s,s} = z$, write the instance $a = b = c = d = s$, $e = g = 1$ of the second hexagon family, and show with Theorem 7.8 that $z = \pm i$. Draw the curl that defines θ_s (Equation (7.7)) and compute θ_s and the double braiding for the semion $z = i$.

(ii) For Fibonacci with $\theta_\tau = e^{4\pi i/5}$, check the ribbon relation $R_c^{\tau\tau}R_c^{\tau\tau} = \theta_c/\theta_\tau^2$ for $c = 1, \tau$. Is τ transparent?

Hint. (i) In the second family inverted crossings carry their labels exchanged; $R_s^{1s} = 1$.

Answer. (i) $z^{-1}\omega z^{-1} = \omega^2$, so $z^2 = \omega = -1$; $\theta_s = z = i$, double braiding $z^2 = -1$. (ii) $(R_1^{\tau\tau})^2 = e^{-8\pi i/5} = \theta_\tau^{-2}$ and $(R_\tau^{\tau\tau})^2 = e^{6\pi i/5} = \theta_\tau^{-1}$; no: neither square is 1.

Exercise 6 ⋆. *The Ising hexagons, solved.* Take Ising with the F-symbols of Theorem 6.7 for $\kappa = +1$ ($F_\sigma^{\sigma\sigma\sigma} = H$, $[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma} = [F_\sigma^{\psi\sigma\psi}]_{\sigma\sigma} = -1$, the rest 1 where the trees exist). The unknowns are $R_1^{\sigma\sigma}$, $R_\psi^{\sigma\sigma}$, $R_\sigma^{\sigma\psi}$, $R_\sigma^{\psi\sigma}$ and $R_1^{\psi\psi}$.

(i) Write out the first- and the second-family instances with labels $(a,b,c,d,e,g) = (\sigma,\psi,\sigma,\psi,1,\sigma)$, and deduce $R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma}$.

(ii) Write out the first-family instances with labels $(\psi,\psi,\sigma,\sigma,\sigma,\sigma)$, $(\sigma,\sigma,\psi,1,\sigma,\sigma)$ and $(\sigma,\sigma,\sigma,\sigma,1,1)$, and use them to show that there are at most four solutions. Which values of ν in $\theta_\sigma = e^{i\pi\nu/8}$ do they give?

(iii) Write out the solution with $\nu = 1$, and compute θ_ψ.

Hint. (i) In each instance check which trees exist; only one f survives. (ii) Start with the first label set; for the spin use Theorem 7.7.

Answer. (i) $R_1^{\sigma\sigma} R_\sigma^{\sigma\psi} = -R_\psi^{\sigma\sigma}$ and $R_1^{\sigma\sigma} R_\sigma^{\psi\sigma} = -R_\psi^{\sigma\sigma}$. (ii) $y = R_\sigma^{\sigma\psi}$ has $y^2 = -1$, $R_1^{\psi\psi} = y^2 = -1$, $\sqrt{2}\,(R_1^{\sigma\sigma})^2 = 1 + y$: $R_1^{\sigma\sigma} = \pm e^{\pm i\pi/8}$ with the exponent's sign that of y/i; $\nu \in \{1, 7, 9, 15\}$. (iii) $R_1^{\sigma\sigma} = e^{-i\pi/8}$, $R_\psi^{\sigma\sigma} = e^{3i\pi/8}$, $R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma} = -i$, $R_1^{\psi\psi} = -1$; $\theta_\psi = -1$.